

5.1 Introduction to Computing Impacts

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 01 of 12

5.1 Introduction to Computing Impacts

LEARN IT

Computing means using computers and computer-based devices to store, process and share information. Today computing is an important part of your daily life. You use computers, mobile phones, the internet and smart devices almost everywhere — at home, in college and on the road.

In Pakistan, computing is becoming **vital** (very necessary) in:

- **Schools and colleges** — online admissions, e-books, online results of the Sindh boards.
- **Hospitals** — computerised patient records, digital X-rays, online appointment systems.
- **Banks** — ATMs, mobile banking apps, Raast instant payments, JazzCash and Easypaisa wallets.
- **Farms** — weather forecasts on mobile phones, soil sensors, tube-well control.
- **Industries** — automatic machines in textile mills in Karachi, Hyderabad and Faisalabad.

An **impact** is the effect something has on people and society. Computing has **positive impacts** (speed, convenience, new jobs) and **negative impacts** (privacy risks, addiction, job loss). In this chapter you will study three key technologies first — AI, IoT and Data Analytics — and then their uses and impacts.

WORKED EXAMPLE

A day in Sara's life in Karachi — spot the computing:

Step 1. 7:00 am — Her phone unlocks by recognising her face (AI).

Step 2. 7:30 am — She books a ride on a ride-hailing app; GPS shows the car's location (internet + location data).

Step 3. 9:00 am — College attendance is marked by a biometric (fingerprint) machine (IoT).

Step 4. 2:00 pm — She pays her fee through a mobile banking app (computing in business).

Step 5. 8:00 pm — YouTube suggests a physics lecture based on what she watched yesterday (AI).

Conclusion: in a single day, Sara uses computing in at least five different fields without even thinking about it.

اردو میں سمجھیں

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔ پاکستان میں اسکول، اسپتال، بینک، کھیت اور صنعتیں سب کمپیوٹر اور انٹرنیٹ پر انحصار کرنے لگے ہیں۔ کمپیوٹنگ کے فائدے بھی ہیں اور نقصانات بھی، جنہیں ہم اس باب میں سمجھیں گے۔

CHECK YOURSELF

1. Name four places in Pakistan where computing has become vital.

2. What is meant by the "impact" of computing?

Answers: 1. Any four of: schools, hospitals, banks, farms, industries. 2. The effect computing has on people and society — it can be positive or negative.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Computing is now part of:

- a. Daily life — phones, banks, schools
- b. Only museums
- c. Only NAND labs
- d. Only OSI textbooks

MCQ 2. An impact can be:

- a. Positive or negative
- b. Only positive
- c. Only hardware heat
- d. Only Big O

MCQ 3. This unit's three big technologies are:

- a. AI, IoT, Data analytics
- b. Only bubble sort
- c. Only Access forms
- d. Only Figma

MCQ 4. A smart meter is closest to:

- a. IoT
- b. A primary key
- c. A K-map
- d. A sprint

MCQ 5. Predicting fail risk from marks is:

- a. Data analytics / AI
- b. Physical cabling
- c. A pie only
- d. Waterfall always

MCQ 6. Students must also learn:

- a. Limits and ethics, not only hype
- b. That computers never fail
- c. That data is always true
- d. That IoT needs no internet

Extra short questions

1. Give two daily computing uses in Pakistan.
2. Why study impacts, not only coding?
3. Name the three tools this chapter compares.
4. One negative impact.

Extra long question

1. Write a balanced 10-mark intro: three benefits of computing in a Sindh college and three harms, with one mitigation each.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. JazzCash/Easypaisa; NADRA records; online boards; WhatsApp classrooms — any two.
2. Technology changes jobs, privacy, inequality — exams ask both sides.
3. AI, IoT, data analytics.
4. Job displacement, addiction, surveillance, e-waste.

Long

1. Benefits: attendance systems, online notes, analytics of results. Harms: distraction, data leaks, unfair access. Mitigations: phone policy, passwords/backups, lab hours for students without devices.

Next lecture: 5.1.1 Artificial Intelligence (AI)